CCC-C-439E
June 2, 1969
SUPERSEDING
Fed. Spec. CCC-C-439d
May 1, 1964

FEDERAL SPECIFICATION

CLOTH, BUNTING, COTTON, MERCERIZED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers one type of cotton bunting.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-P-1134 - Packaging and Packing of Cotton and Cotton-Synthetic Fiber Blend Fabrics (Excluding Duck Fabrics).

Federal Standards:

Fed. Std. No. 4 - Glossary of Fabric Imperfections.
Fed. Test Method Std. No. 191 - Textile Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from the Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, San Francisco, Los Angeles and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:29Z
Check the source to verify that this is the current version before use.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Color Association of the United States, Inc.
United States Army Color Card for Official Colors for Army and Services.

(Application for copies should be addressed to Color Association of the United States, Inc., 200 Madison Ave., New York, N.Y. 10016.)

Rules and Regulations Under the Textile Fiber Products Identification Act.

(Copies may be obtained without charge from the Federal Trade Commission, Washington, D.C. 20580.)

3. REQUIREMENTS

3.1 Standard Sample. When a standard sample is available, the finished cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 Material.

3.2.1 Yarn. The yarn shall be made from cotton which has been carded, drawn and spun into two-ply yarn for both warp and filling.

3.3 Color. The color shall be as specified (see 6.2) and shall match the standard sample. When a standard sample is not available the color shall match the applicable cable number of the Color Card of the Color Association of the United States, Inc. (see 2.2) or color number specified as referenced in any other commercial Color Index. The use of dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid is prohibited. The dyestuff shall contain no more labile sulfur than shown by the standard sample (see 6.3) when tested as specified in 4.4. When a standard sample is not available the dyed and finished cloth shall show no more than a slight trace of labile sulfur as defined in the test method when tested as specified in 4.4.

3.3.1 Matching. The dyed and finished cloth shall match the standard sample or the specified cable number under natural (north sky) daylight or artificial daylight having color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

3.3.2 Colorfastness. The dyed and finished cloth shall show colorfastness to crocking, light, water, and salt water and soap equal to or better than the standard sample when tested as specified in 4.4. When no standard sample is established, the dyed and finished cloth shall show "fair" colorfastness to crocking, and "good" colorfastness to light, water and salt water and soap when tested as specified in 4.4.

3.4 Physical requirements. The finished cloth shall conform to the physical requirements specified in table I when tested as specified in 4.4.

TABLE I. Physical requirements

Weight per square yard ounces	Yarns per inch (minimum)		Breaking strength pounds (minimum)	
	Warp	Filling	Warp	Filling
4.5	31	30	50	50

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:29Z
Check the source to verify that this is the current version before use.

3.4.1 Weave. The weave shall be plain.

3.4.2 Width. The minimum width exclusive of selvages shall be 37-1/2 inches for the Military services and 35-3/4 inches for Civil agencies (see 6.2).

3.5 Finish. The cloth shall be mercerized.

3.6 Nonfibrous materials. The starch and protein content including chloroform-soluble and water-soluble material shall not exceed 2.0 percent when tested as specified in 4.4.

3.7 pH. The pH value of the water extract of the dyed and finished cloth shall be no less than 5.0 nor more than 8.5 when tested as specified in 4.4.

3.8 Length and put-up. Unless otherwise specified (see 6.2), the cloth shall be in continuous pieces of no less than 35 yards and shall be put-up in rolls or flat folded pieces as specified in PPP-P-1134.

3.9 Fiber identification. Each roll or folded piece shall be labeled, ticketed or invoiced for fiber content in accordance with the Textile Fiber Products Identification Act.

3.10 Marking. The face side of the cloth shall be identified by applying a stamping on that side of the cloth with the word "face" at each end of the piece.

3.11 Workmanship. The cloth shall conform to the quality established by this specification. The demerit points per 100 square yards when calculated as specified in section 4 shall not exceed the established maximum point value.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliances are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.2.2 Examination of the end item. Examination of the end item shall be in accordance with 4.2.2.1 through 4.2.2.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:29Z
Check the source to verify that this is the current version before use.

4.2.2.1 Yard-by-yard examination. Each roll or flat folded piece in the sample shall be examined on the face side only. When the total yardage in the roll or flat folded piece does not exceed 100 yards, the entire yardage in the roll or flat folded piece shall be examined. When the total yardage in the roll or flat folded piece exceeds 100 yards, only 100 yards shall be examined. All defects as defined in Section I of Fed. Std. No. 4, which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.2.2.1.1 except that only those slubs and knots which exceed the maximum limits shown for each in figure 1 of Fed. Std. No. 4 shall be scored, and coarse yarns and jerked-in filling, glossary numbers 2, 3 and 26 shall not be scored. No linear yard (folded yards, or increments of one yard on the measuring device of the inspection machine) from any one piece within the sample shall be penalized more than four points. The sample size shall be 20 rolls, or flat-folded pieces, selected from 20 containers. The lot shall be unacceptable if the points per 100 square yards of the total yardage examined exceeds 55 points. The lot shall be unacceptable if the points per 100 square yards of two or more individual rolls or folded pieces exceeds 83 points. If no individual roll or folded piece exceeds 83 points per 100 square yards, the lot shall be acceptable with respect to this characteristic. If one roll or flat folded piece exceeds 83 points per 100 square yards, a second sample of 20 rolls shall be examined only for individual roll quality examination. The lot shall be unacceptable if one or more rolls or flat folded pieces in the second sample exceeds 83 points per 100 square yards. Point computation for lot quality and individual roll or piece quality shall be as follows:

$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{total yards inspected}} = \text{Points per 100 square yards}$$

4.2.2.1.1 Demerit points. Demerit points shall be assigned as follows:

For defects up to 3 inches in any dimension	- one point
For defects exceeding 3 inches, but not exceeding 6 inches in any dimension	- two points
For defects exceeding 6 inches, but not exceeding 9 inches in any dimension	- three points
For defects exceeding 9 inches in any dimension	- four points

The following defects, when present, shall be scored four points for each yard in which they occur:

Baggy, ridgy, or wavy cloth.
Objectionable odor.
Width less than specified.
Poor dye penetration, mottled, streaky or cloudy.

4.2.2.2 Examination for length.

4.2.2.2.1 Individual pieces. During the yard-by-yard examination, each roll or flat folded piece in the sample shall be examined for length. Any length found to be less than the minimum specified or more than two yards less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls or flat folded pieces in the sample are defective in respect to length.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:29Z
Check the source to verify that this is the current version before use.

4.2.2.2.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls or flat folded pieces in the sample is less than the total of the lengths marked on the ticket.

4.2.2.3 Examination for shade. During the yard-by-yard examinations, each roll or flat folded piece in the sample shall be examined for shade. Any roll or flat folded piece in the sample off shade, shaded side to side, side to center or end to end shall be cause for rejection of the entire lot represented by the sample.

4.2.2.4 Examination for face markings and compliance with the Textile Fiber Products Identification Act. During the yard-by-yard examination, each roll or flat folded piece in the sample shall be examined for these defects. The lot shall be unacceptable if two or more rolls or flat folded pieces in the sample have the face stamping missing from either or both ends, or the face stamping on the wrong side; or are not labeled, ticketed, or invoiced in accordance with the Textile Fiber Products Identification Act.

4.3 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisions of PPP-P-1134 to determine whether packaging, packing and marking requirements of section 5 of this specification are complied with.

4.4 Test methods. The methods of testing specified in Fed. Std. No. 191, wherever applicable, and as listed in table II shall be followed. The physical and chemical values specified in section 3, except where otherwise specified, apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test method. All test reports shall contain the individual values utilized in expressing the final result. The sample unit for test purposes shall be 1-1/2 continuous yards full width of the finished cloth. The lot shall be unacceptable if one or more units fail to meet any requirement specified. The sample size shall be as follows:

Lot size (yards)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE II. Test methods

Characteristic	Requirement paragraph	Test method
Identification of cotton	3.2.1	1200 1/
Yarn ply	3.2.1	Visual 2/
Presence of labile sulfur	3.3	2020
Colorfastness to:		
Crocking	3.3.2	5651
Light	3.3.2	5660
Water	3.3.2	5630
Salt water and soap	3.3.2	5632
Weight	3.4	5041
Yarns per inch:		
Warp	3.4	5050
Filling	3.4	5050
Breaking strength:		
Warp	3.4	5100
Filling	3.4	5100
Weave	3.4.1	Visual 2/
Mercerization	3.5	Microscopic examination for swelling of cotton fibers 1/
Nonfibrous material	3.6	2611
pH	3.7	2811

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:29Z
Check the source to verify that this is the current version before use.

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ One determination shall be made from each sample unit and the results reported as "pass" or "fail".

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2 and 6.2.1).

5.1.1 Levels A and C. The cloth, put up as specified, shall be packaged in accordance with the applicable requirements of PPP-P-1134.

5.2 Packing.. Packing shall be level A, B or C as specified (see 6.2 and 6.2.1).

5.2.1 Levels A, B and C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1134.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with PPP-P-1134.

6. NOTES

6.1 Intended use. The bunting is primarily intended for use in the manufacture of signal, semaphore, ensign (including U.S. flag) and other type flags.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) Color required (see 3.3).
(c) Width required (see 3.4.2).
(d) When length of piece is other than specified (see 3.8).
(e) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2.1 In the preparation of contracts or orders, it must be noted that various options, choices and alternatives, as indicated in PPP-P-1134, may be exercised in the preparation for delivery of this commodity.

6.3 Standard sample. For access to standard sample, address the procuring office issuing the invitation for bids.

6.4 Dyestuff formulations. The following are suggested dyestuff formulations for some Red, Yellow, Blue and Black colors:

Old Glory Red 70180

Azoic Coupling 2, C.I. 37505
Azoic Coupling 29, C.I. 37527
Azoic Diazo 9, C.I. 37040

Old Glory Blue 70075

Vat Blue 14, C.I. 69810
Vat Violet 13, C.I. 68700
Vat Violet 9, C.I. 60005
Vat Blue 20, C.I. 59800
Vat Black 27, C.I. 69005

Yellow 65002

Vat Yellow 22, No C.I. number

Black 65018

Vat Black 9, C.I. 65230
Vat Black 25, C.I. 69525

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:29Z
Check the source to verify that this is the current version before use.

6.5 For civil agency procurement. When level B packaging is required for civil agency procurement, the level A packaging requirements shall apply.

MILITARY CUSTODIANS:

Army - GL
Navy - SA
Air Force - 82

Review activities:

Army - IH, MD
Navy - SA
Air Force - 11, 82, 85

User activity:

Navy - MC

Preparing activity:

Army - GL

CIVIL AGENCIES INTEREST:

COM-NBS
GSA-FSS
JUS-FPI
VA-DMS

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 10 cents each.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:29Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

Cloth, Bunting, Cotton, Mercerized CCC-C-439E

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:29Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY
U. S. Army Natick Laboratories
Natick, Massachusetts 01760

OFFICIAL BUSINESS

POSTAGE & FEES PAID
DEPARTMENT OF THE ARMY

Commanding General
U. S. Army Natick Laboratories
ATTN: AMXRES-EQS
Natick, Massachusetts 01760

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:29Z
Check the source to verify that this is the current version before use.